

Figure 2.15 Cardinality limits on relationship sets.

2.6 Weak Entity Sets

An entity set may not have sufficient attributes to form a primary key. Such an entity set is termed a **weak entity set**. An entity set that has a primary key is termed a **strong entity set**.

As an illustration, consider the entity set *payment*, which has the three attributes: *payment-number*, *payment-date*, and *payment-amount*. Payment numbers are typically sequential numbers, starting from 1, generated separately for each loan. Thus, although each *payment* entity is distinct, payments for different loans may share the same payment number. Thus, this entity set does not have a primary key; it is a weak entity set.

For a weak entity set to be meaningful, it must be associated with another entity set, called the **identifying** or **owner entity set**. Every weak entity must be associated with an identifying entity; that is, the weak entity set is said to be **existence dependent** on the identifying entity set. The identifying entity set is said to **own** the weak entity set that it identifies. The relationship associating the weak entity set with the identifying entity set is called the **identifying relationship**. The identifying relationship is many to one from the weak entity set to the identifying entity set, and the participation of the weak entity set in the relationship is total.

In our example, the identifying entity set for *payment* is *loan*, and a relationship *loan-payment* that associates *payment* entities with their corresponding *loan* entities is the identifying relationship.

Although a weak entity set does not have a primary key, we nevertheless need a means of distinguishing among all those entities in the weak entity set that depend on one particular strong entity. The **discriminator** of a weak entity set is a set of attributes that allows this distinction to be made. For example, the discriminator of the weak entity set *payment* is the attribute *payment-number*, since, for each loan, a payment number uniquely identifies one single payment for that loan. The discriminator of a weak entity set is also called the *partial key* of the entity set.

The primary key of a weak entity set is formed by the primary key of the identifying entity set, plus the weak entity set's discriminator. In the case of the entity set *payment*, its primary key is {*loan-number*, *payment-number*}, where *loan-number* is the primary key of the identifying entity set, namely *loan*, and *payment-number* distinguishes *payment* entities within the same loan.

The identifying relationship set should have no descriptive attributes, since any required attributes can be associated with the weak entity set (see the discussion of moving relationship-set attributes to participating entity sets in Section 2.2.1).

A weak entity set can participate in relationships other than the identifying relationship. For instance, the *payment* entity could participate in a relationship with the *account* entity set, identifying the account from which the payment was made. A weak entity set may participate as owner in an identifying relationship with another weak entity set. It is also possible to have a weak entity set with more than one identifying entity set. A particular weak entity would then be identified by a combination of entities, one from each identifying entity set. The primary key of the weak entity set would consist of the union of the primary keys of the identifying entity sets, plus the discriminator of the weak entity set.

In E-R diagrams, a doubly outlined box indicates a weak entity set, and a doubly outlined diamond indicates the corresponding identifying relationship. In Figure 2.16, the weak entity set *payment* depends on the strong entity set *loan* via the relationship set *loan-payment*.

The figure also illustrates the use of double lines to indicate *total participation*—the participation of the (weak) entity set *payment* in the relationship *loan-payment* is total, meaning that every payment must be related via *loan-payment* to some loan. Finally, the arrow from *loan-payment* to *loan* indicates that each payment is for a single loan. The discriminator of a weak entity set also is underlined, but with a dashed, rather than a solid, line.

In some cases, the database designer may choose to express a weak entity set as a multivalued composite attribute of the owner entity set. In our example, this alternative would require that the entity set *loan* have a multivalued, composite attribute *payment*, consisting of *payment-number*, *payment-date*, and *payment-amount*. A weak entity set may be more appropriately modeled as an attribute if it participates in only the identifying relationship, and if it has few attributes. Conversely, a weak-entity-set representation will more aptly model a situation where the set participates in relationships other than the identifying relationship, and where the weak entity set has several attributes.

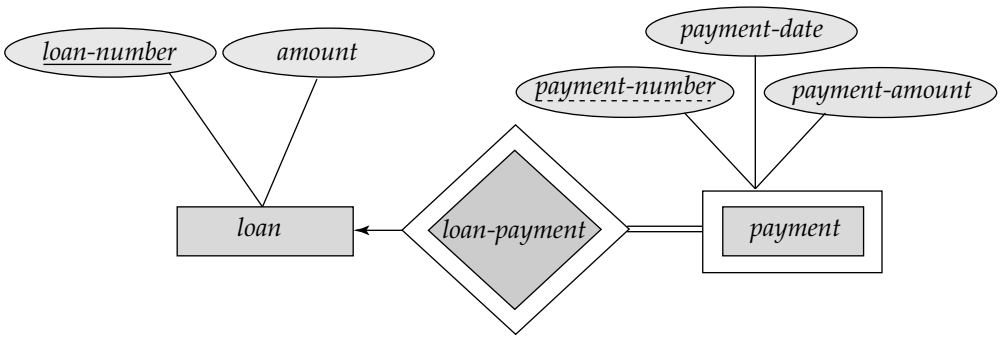


Figure 2.16 E-R diagram with a weak entity set.